

PROJECT TITLE: OptiCrop: Smart Agricultural Production Optimization Engine

1. IDEA GENERATION (BRAINSTORMING):

During the initial project ideation phase, our team explored multiple computational approaches to resolve the crop yield optimization problem:

- * Concept A: Automated IoT-based live sensor deployment across fields. (Rejected due to high hardware costs for poor farmers).
- * Concept B: Deep Learning image analysis of soil surface textures. (Rejected due to poor accuracy variations across differing weather shades).
- * Concept C: Data-driven Predictive Analysis Framework based on historical soil metrics (N, P, K, pH) and localized climate averages (Temperature, Humidity, Rainfall). (Selected and Prioritized).

2. IDEA PRIORITIZATION & SCOPE MODIFICATION:

We deliberately focused on a simplified, accessible, and high-impact scope. Instead of demanding complex, highly expensive, and inaccessible laboratory soil reports from farmers, OptiCrop relies on standard direct numbers for crop selection parameters. The target priorities are sorted based on:

- * Feasibility: Running lightweight Machine Learning algorithms (Random Forest) on minimal datasets.
- * Accessibility: A simple web-based frontend requiring no installation.
- * Performance: Instant recommendations under 2 seconds.